

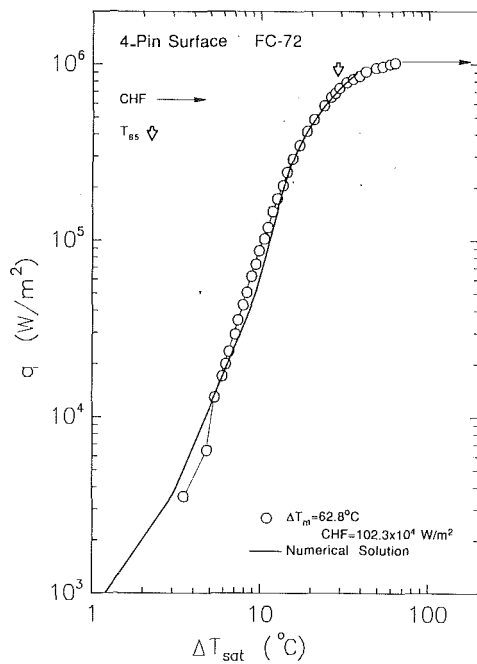


Fig. 10(a)

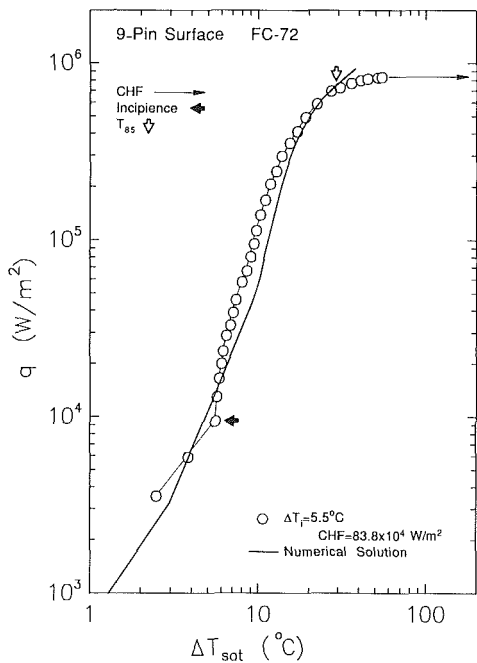
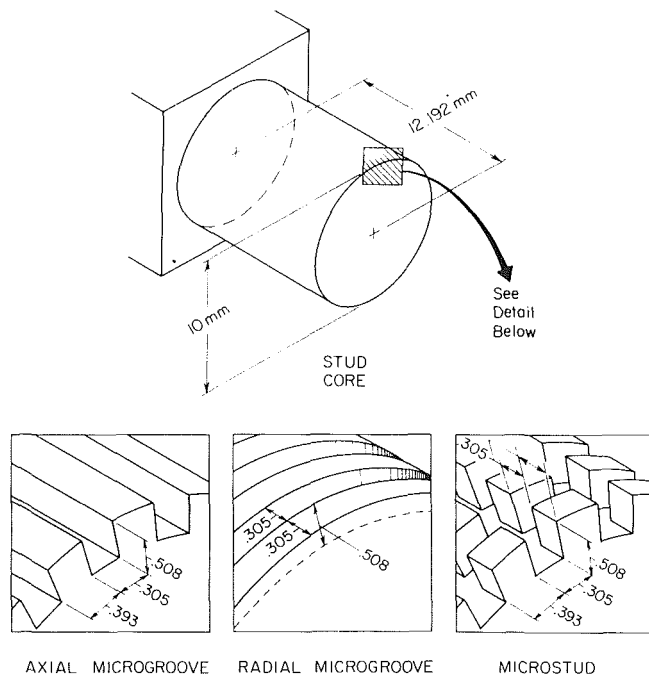


Fig. 10(b)

Fig. 10 Boiling curves for the (a) 4-pin and (b) 9-pin surfaces in saturated FC-72

sity regions of the boiling curve (Anderson and Mudawar, 1989). The machined microstructures, on the other hand, act as active nucleation sites during more developed boiling since the characteristic fin and gap widths for these microstructures are smaller than a typical bubble departure diameter. The diameter of the studs was roughly two times greater than the Taylor wavelength, λ_T (5.87 mm for FC-72) and approximately equal to the Taylor "most dangerous" wavelength $\lambda_D (= \sqrt{3}\lambda_T)$ associated with CHF. Therefore, it was expected that CHF would be accompanied by the formation of a single vapor blanket in the cross-wise plane of these fins.

Data for the microgroove-enhanced surfaces in saturated FC-72 are shown in Fig. 12(a). In the natural convection region,



AXIAL MICROGROOVE RADIAL MICROGROOVE MICROSTUD

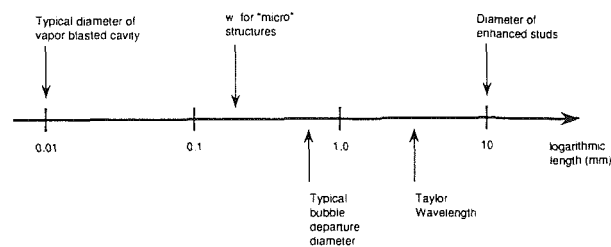


Fig. 11 Geometries of the enhanced stud surface and associated characteristic length scales

the axially-grooved stud turbulent the induced liquid flow more effectively than the radially-grooved stud as evidenced by the higher natural convection heat transfer rate for the axially-grooved stud. Numerically, the distinction between the direction of the grooves is not taken into consideration; hence, the predicted boiling curves are equivalent for both structures, based on the low-profile microgroove data provided in Fig. 3 and Table 2(a). Figure 12(b) shows a slight discrepancy between numerical predictions and experimental data for the microstud-enhanced stud based on the reference low-profile microstud data given in Fig. 3 and Table 2(b), especially in the low heat flux region. This surface achieved a CHF value of 105.4 W/cm² in saturated FC-72, the highest of all the enhanced surface geometries.

Experiments were performed to combine the attractive features of subcooled boiling with the high CHF values of the enhanced-stud surfaces. Comparisons of saturated and subcooled boiling curves for the low profile microstud reference surface, Fig. 3, and microstud-enhanced stud, Fig. 11, are shown in Figs. 13(a) and 13(b), respectively. Figure 13(a) shows how subcooling eliminated the incipience excursion associated with saturated boiling on the reference microstud surface and increased CHF by 42 percent. The numerical results for subcooled boiling show good agreement with the microstud-enhanced stud, Fig. 13(b), like it did for saturated boiling. The same surface produced the highest heat flux in the present study with subcooled boiling. The 159.3 W/cm² heat dissipation capability of this surface suggests pool boiling is a viable alternative for cooling even very high power electronic chips.